

TRACE: Transformer for Regularized and Accurate Cortical ECoG Motor Decoding

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Abstract—Electrocorticography (ECoG) records local field potentials from the cortical surface with spatial resolution and signal stability that non-invasive modalities such as scalp electroencephalograms (EEGs) cannot match, making it a strong candidate for motor brain-computer interfaces (BCIs). Convolutional architectures have driven recent advances in ECoG motor decoding, but their reliance on local receptive fields limits their ability to capture long-range temporal dependencies in continuous motor signals. We propose Transformer for Regularized and Accurate Cortical ECoG Motor Decoding (TRACE), a transformer-based architecture that applies multi-head self-attention across ECoG electrode channels and time to model distributed spatiotemporal patterns of cortical motor activity. On the BCI Competition IV Dataset 4, TRACE achieves a new state of the art mean Pearson correlation of 0.724 across subjects, exceeding both FingerFlex (0.597), the widely-used open-source U-Net baseline from 2023, and DTCNet (0.680), the previous state of the art from 2025. Ablation studies identify severe channel dropout ($p=0.6$), a form of input-level regularization that forces the model to decode from distributed cortical representations, as the single most impactful design choice, with direct implications for clinical deployment on variable-coverage grids. On a nine-patient finger-flexion subset of the Miller ECoG Library, TRACE achieves a new state of the art mean Pearson correlation of 0.508, up from U-Net’s performance of 0.436.

Index Terms—electrocorticography, brain-computer interface, motor decoding, transformer, attention mechanism, deep learning

I. INTRODUCTION

More than 5 million people in the United States live with some form of paralysis [1], and upper limb weakness persists in more than three-quarters of stroke survivors [2], [3], [4]. Motor BCIs that directly interpret neural signals into control commands for robotic arms, cursors, and prosthetic devices represent a vital component of treatment and recovery for these patients, offering a path to restored motor function where conventional rehabilitation plateaus [5]. Electrocorticography (ECoG) electrode grids placed on the cortical surface record local field potentials at 1 cm inter-electrode spacing with high signal-to-noise ratio [6], and power modulations in the gamma and high-gamma bands (40–300 Hz) track ongoing finger and hand movements with sufficient fidelity for trajectory-level decoding [7], [8]. Convolutional architectures remain the dominant approach for decoding continuous finger movements from ECoG, but their reliance on local receptive fields limits their capacity to model long-range temporal dependencies in

motor signals [9], [10]. The original evaluation of the U-Net-inspired FingerFlex architecture [11], for example, achieved a mean Pearson correlation of 0.67 across three subjects on the BCI Competition IV Dataset 4 [8], a substantial improvement over the best result of 0.48 from the original competition and the standard open-source benchmark for this task.

The transformer architecture [12] has been widely adopted in scalp electroencephalogram (EEG) motor imagery classification [10], [13]. However, it has not been systematically evaluated on ECoG data: no prior study has benchmarked attention-based architectures against convolutional baselines across any ECoG-based task. This gap leaves open the question of whether transformers provide meaningful gains over Convolutional Neural Networks (CNNs) for continuous ECoG motor decoding, and whether such gains generalize across subjects and task types.

This paper addresses this gap by introducing Transformer for Regularized and Accurate Cortical ECoG Motor Decoding (TRACE), a transformer architecture that applies multi-head self-attention across ECoG electrode channels and time to capture distributed spatiotemporal patterns of cortical motor activity. On BCI Competition IV Dataset 4, TRACE achieves a mean Pearson correlation of 0.724 across subjects, outperforming a U-Net FingerFlex baseline (0.597) and previous state-of-the-art DTCNet [14] (0.680) under a unified evaluation protocol applied to all baselines. We identify channel dropout as the dominant regularizer on this benchmark: optimizing the per-step electrode drop rate from $p=0.1$ to an ideal value of $p=0.6$ yields a mean correlation gain of +0.079. On the nine-patient Miller ECoG Library finger-flexion subset, where clinically placed grids make decoding substantially harder, TRACE achieves a mean Pearson correlation of 0.508, up from the U-Net baseline’s 0.436.

II. RELATED WORK

A. Finger Trajectory Decoding on BCI Competition IV Dataset 4

BCI Competition IV Dataset 4 remains the standard public benchmark for continuous finger flexion prediction from human ECoG [8]. The dataset consists of 10-minute ECoG recordings from three epileptic patients with the corresponding finger movements. Finger flexion percentages, where 0% indicates fully extended and 100% indicates fully curled, and tim-

ing were recorded via instrumented data gloves. Each subject was instructed to move specific fingers in response to visual cues displayed on a computer monitor. The recording grids consisted of 48–64 electrodes, but the creators of the datasets scrambled electrode positions relative to brain anatomy to preserve patient privacy.

Early competition entries established linear and structured baselines. Liang and Bougrain [15] won the original competition with a linear regression method using amplitude modulation of band-specific ECoG features with autoregressive short-term memory terms for individual finger flexion prediction. Flamary and Rakotomamonjy [16] proposed switching linear models controlled by a hidden state that first estimates which finger is moving and then predicts flexion; their entry placed second with a mean correlation of approximately 0.42 across fingers and subjects on the official test split.

Deep learning has since substantially raised reported correlations on the same benchmark. FingerFlex [11] adapts a U-Net [17]-style convolutional encoder-decoder with skip connections for trajectory regression, achieving a mean correlation of 0.67 in its initial evaluation on the BCI Competition 4 dataset—higher than all previous methods. Ablation studies highlighted data scaling, architectural refinements, and skip connections as key contributors to performance. Wang et al. [14] proposed DTCNet, a compact CNN-attention hybrid that preserves the 3D structure of wavelet spectrograms (channels $\times$ frequencies $\times$ time) and uses dilated-transposed convolutions to capture temporal dependencies at multiple scales. DTCNet reported a mean correlation of 0.69 on Dataset 4, which we reproduce at 0.680, making it the strongest independently verified baseline prior to TRACE. Jangir et al. [18] reported a substantially higher mean correlation of 0.85 with BC4D4, a hybrid CNN-DNN pipeline, but their evaluation removes approximately 70% of samples via outlier filtering, making direct comparison or reproduction difficult. Together, these lines of work show that architectural inductive bias, preprocessing, and evaluation protocol all materially affect reported scores on this fixed benchmark.

B. Miller ECoG Library

“A Library of Human Electrographic Data and Analyses” [19], compiled by Kai J. Miller at Stanford University and commonly referred to as the Miller ECoG Library, provides a broader and more clinically representative dataset. The Miller library includes both motor and non-motor tasks across dozens of patients, of which finger flexion is one subset. It is relatively new as a deep-learning benchmark and substantially harder than Dataset 4: electrode placement follows clinical seizure-localization needs instead of BCI-optimized targeting of hand sensorimotor cortex, so coverage of informative motor areas ranges from dense to marginal across patients. Per-subject training data are shorter and less standardized. Finger-flexion labels are comparatively noisier than in Dataset 4’s tightly cued paradigm. These factors cap the achievable correlation well below Dataset 4 and inflate cross-subject variance. The open-source FingerFlex [11] reports per-patient

correlations on nine Miller subjects, averaging roughly 0.49 with a range of 0.22–0.71 that reflects the variety of grid coverage. Tao et al. [20] proposed DeepFingerNet, a nested U-Net architecture, and reported a mean correlation of 0.54 on the same subset; however, we were unable to reproduce these results despite multiple attempts matching the published architecture and preprocessing (Section V-D). Evaluating on the Miller Library alongside Dataset 4 tests whether decoder gains transfer to the sparse, clinically placed grids that characterize real epilepsy-monitoring scenarios, a necessary step toward clinical translation.

C. Transformers for Neural Time Series

The Transformer architecture [12] and its variants have become a default tool for sequence modeling in many domains. For non-invasive EEG, Song et al. [10] introduced EEG Conformer, a convolutional transformer that combines local feature extraction with global self-attention, and reported gains on standard motor-imagery benchmarks. Xie et al. [13] described CNN-Transformer hybrids in which a CNN performs spatial and temporal feature extraction before features are embedded with positional encoding and processed by transformer blocks; a fused model processes spatial and temporal streams in parallel before combining them. These results motivate evaluating whether similar attention-based modeling transfers to invasive ECoG trajectory regression, where labels are continuous and the spatial layout of electrodes is anonymized in Dataset 4.

D. Reproducibility and Benchmarking in Motor BCI

Comparing ECoG decoding results across studies is difficult in practice because evaluation protocols vary in ways that are rarely documented. EEG-based motor imagery work reports classification accuracy on discrete trial labels, while ECoG trajectory studies report Pearson correlation on continuous outputs, and neither metric translates directly to the other. A model that achieves 0.70 correlation on one benchmark cannot be ranked against a model reporting 92% accuracy on a different task without shared evaluation criteria.

The problem extends beyond metrics. Most published ECoG studies train and evaluate on institution-specific datasets with between 9 and 21 subjects, and electrode placements, grid sizes, and task protocols differ enough across sites that pooling results is not straightforward. PhysioNet [21], with 109 subjects, is one of the few public datasets with sufficient scale for cross-subject generalization analysis, but it provides scalp EEG rather than ECoG, leaving invasive motor decoding benchmarks fragmented [8].

Given the ECoG benchmark inconsistencies, independently reproducing published results by reimplementing architectures, verifying preprocessing, and evaluating under a unified protocol is essential before treating reported correlations on Dataset 4 as ground truth. Source code availability did not determine our choice of baselines as we reimplemented every architecture compared in this paper from the published description regardless of whether code was released. However, it did affect the

ease and confidence of reproduction. Among the ECoG finger-decoding methods surveyed here, FingerFlex was the only model with a public code release, and our experience reproducing closed-source architectures (Section V-D) suggests that broader open-source practice would substantially accelerate progress and reduce wasted replication effort in this field.

III. DATA AND PREPROCESSING

A. Datasets

We evaluate models on BCI Competition IV Dataset 4 [8] (3 subjects, 48–64 channels, ~10 min each, five-finger flexion labels from instrumented gloves) and a nine-patient finger-flexion subset of the Miller ECoG Library [19] (38–64 channels, 179–610 s per patient). Together these span twelve subjects. Each subject has its own trained model and no data is shared across subjects.

B. Preprocessing Pipeline

We adopt the preprocessing pipeline established by FingerFlex [11] for Dataset 4 decoding, with one addition (60 Hz notch filtering). A 60 Hz notch filter removes power-line interference, which we confirmed via PSD analysis of the raw recordings. Bandpass filtering then restricts each channel to the 40–300 Hz band, where ECoG power modulations track ongoing movement [7]. Common average referencing (CAR) subtracts the instantaneous mean across electrodes on each sample, attenuating volume-conducted components shared across the grid. Complex Morlet wavelets decompose each channel into time-frequency representations, and frequency bands are aggregated into channel-wise features suitable for encoder input. Finally, normalizing by median and interquartile range per feature reduces sensitivity to heavy-tailed artifacts relative to variance-based scaling. Finger trajectories are min-max normalized per subject and channel where required by the baseline implementations we compare against.

C. Train/Validation/Test Splits and Evaluation Protocol

Continuous ECoG and kinematics are strongly autocorrelated; random frame-wise shuffling would leak future movement into training batches. We therefore reserve the last 15% of each subject’s official training recording as a contiguous validation tail, after any reaction-time alignment used in the benchmark. The competition held-out test set is never used for hyperparameter search or early stopping. This protocol approximates prospective evaluation on later segments of the same session while respecting the organizer-defined train/test boundary for Dataset 4. Miller Library subjects follow the same within-subject temporal carving for validation, with held-out segments or sessions as provided by the release documentation.

Following FingerFlex-style evaluation [11], we optionally apply one-dimensional Gaussian smoothing along time to model outputs before computing Pearson correlation between predicted and recorded flexion. Two settings matter for interpretation. $\sigma=6$ produces smoother trajectories and typically increases reported correlation relative to raw predictions; it

aligns with published validation-style numbers commonly quoted for convolutional baselines on Dataset 4 and eases comparison to those references. $\sigma=1$ applies a milder kernel and yields a more conservative correlation that is closer to the unsmoothed mismatch between decoder output and glove traces, which is relevant when judging deployable decoding quality. Because the correlations are broadly similar across the two settings, reporting in the literature is mixed; we encourage future studies to adopt $\sigma=1$ as the default to better reflect real-time decoding fidelity.

IV. TRACE ARCHITECTURE

TRACE is our proposed transformer encoder for continuous finger flexion regression from ECoG. It is evaluated against FingerFlex [11], the U-Net-style convolutional baseline described in Section II, which serves as the primary convolutional reference on Dataset 4.

TRACE stacks L transformer encoder blocks on top of a linear projection that embeds concatenated spectral features per time step (and, where used, per-channel tokens) into model dimension d_{model} . Each block applies multi-head self-attention, a position-wise feed-forward network, and residual paths (Fig. 2). We adopt pre-LayerNorm (LayerNorm before attention and before the feed-forward sublayers) because it typically improves optimization stability in deeper transformers compared with the original post-norm ordering [22]. Bidirectional self-attention is used within each evaluation window: for offline trajectory regression on Dataset 4, future and past samples inside the window are both available to every output timestep, so causal masking is unnecessary and would discard information that improves continuous flexion estimates. For low-latency online decoding, a causal-masking variant with right-edge-only prediction would be used instead, at some cost in correlation; this is beyond the scope of this work. Sinusoidal positional encodings sum with token embeddings so that attention can disambiguate absolute time order inside the window, following the sinusoidal positional-encoding scheme introduced in [12] and adopted in transformer-based EEG motor decoding [13], avoiding a learned position table for every offset along long recordings. The encoding for position pos and dimension i is:

$$\text{PE}_{(\text{pos}, 2i)} = \sin\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right), \quad (1)$$

$$\text{PE}_{(\text{pos}, 2i+1)} = \cos\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right). \quad (2)$$

Within each block, the input sequence $X \in \mathbb{R}^{T \times d_{\text{model}}}$ is first normalized, then projected into queries, keys, and values for each of the h attention heads:

$$\text{head}_i = \text{Attention}(XW_i^Q, XW_i^K, XW_i^V), \quad (3)$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (4)$$

where $d_k = d_{\text{model}}/h$ is the per-head key dimension and $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_k}$ are learned projections. The h heads are concatenated and linearly mixed:

$$\text{MultiHead}(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O. \quad (5)$$

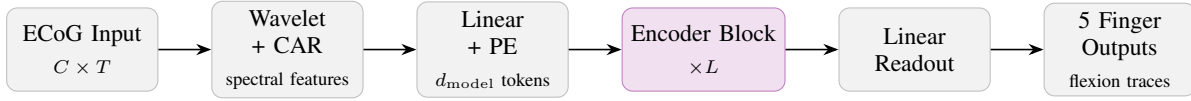


Fig. 1. Overall TRACE pipeline. Raw ECoG (C channels, T samples) is decomposed into time-frequency representations via complex Morlet wavelets with CAR referencing, projected to d_{model} -dimensional tokens with sinusoidal positional encodings (Linear+PE), processed by L stacked encoder blocks (see Fig. 2), and decoded to five continuous finger flexion traces by a linear readout.

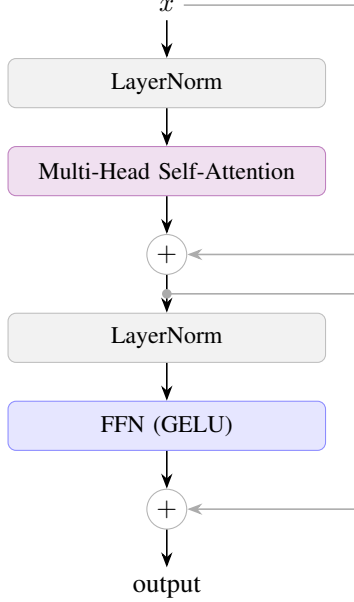


Fig. 2. Pre-LayerNorm encoder block in TRACE, repeated L times. Gray paths are residual connections; circles are element-wise sums.

The $1/\sqrt{d_k}$ scaling prevents dot products from entering the flat region of softmax as d_k grows. The position-wise feed-forward sublayer applies two linear transformations with a GELU nonlinearity [23] between them with an expansion ratio of $4 \times d_{\text{model}}$, followed by a residual connection and layer normalization in pre-norm ordering. Dropout ($p=0.1$) is applied after the positional encoding, after the attention output, and after the FFN output within each block.

Full-session self-attention over all samples of a Dataset 4 run is usually infeasible: memory and compute scale quadratically with sequence length, while ten-minute high-rate ECoG yields very long sequences. Sliding-window inference is therefore *mandatory* for practical training and evaluation: we partition each recording into overlapping windows of fixed length, run the encoder on each window, and stitch or align window-level predictions back to the full timeline for correlation against glove traces. This protocol mirrors how deployed decoders must operate under buffer limits and latency budgets. We use a window of 256 samples at 100 Hz (2.56 s) with a training stride of 1 sample. At inference, a stride of 128 samples (50% overlap) is applied and overlapping predictions are averaged. Studies that rely on localized temporal segments rather than full-sequence models are difficult to compare directly unless window length, stride, and overlap are reported

explicitly.

V. EXPERIMENTS

A. Training Setup

All models are trained within subject on the preprocessed tensors from Section III. Optimization uses Adam with learning rate 8.42×10^{-5} and weight decay 10^{-6} , with ReduceLROnPlateau (factor 0.5). Batch size and stopping criteria vary slightly by architecture. Gradient clipping is applied at global norm 1.0. Mini-batches concatenate random training windows (fixed length and stride as in Section III); batch size is chosen per architecture to fit GPU memory (32–128 windows) and held constant across seeds and subjects.

All models are trained with the hybrid objective $\mathcal{L} = 0.5 \text{MSE} + 0.5(1 - \cos)$ introduced by FingerFlex [11], where MSE is the mean-squared error between predicted and recorded flexion and $\cos$ is the cosine similarity between the predicted and recorded trajectories; this mixture explicitly encourages alignment under correlation-based evaluation. We monitor mean validation Pearson r across five fingers, stopping if no improvement is observed for 20 epochs; the checkpoint with best validation r is evaluated once on the held-out competition test split. Ten random seeds are used for weight initialization, window shuffling, and dropout; reported test scores for TRACE are the mean $\pm$ standard deviation across these seeds on Dataset 4.

B. Transformer Scaling Study

We ablate TRACE along two axes: model capacity (width and depth) and channel dropout rate p . The dropout rate represents the probability that each electrode channel is independently zeroed at the encoder input during training. High dropout rates force the decoder to reconstruct flexion trajectories from an arbitrary subset (of size $1-p$) of the available electrodes on every step, which both regularizes against single-channel reliance and mirrors the clinical reality of variable electrode coverage and signal degradation across patients [19]. Table I summarizes both sweeps on Dataset 4.

Width scaling from $d_{\text{model}}=128$ to 1024 at fixed $p=0.2$ raises mean r from 0.582 to 0.675 (panel a). Per-parameter returns diminish sharply: the first four configurations (0.9–5.9M parameters) add +0.030 mean r in aggregate, while the jump to the 78M production configuration adds a further +0.063 at roughly $13\times$ the parameter cost.

At the production configuration ($d_{\text{model}}=1024$, $L=6$, $h=16$, 78M parameters), sweeping channel dropout from $p=0.1$ to $p=0.6$ adds a further +0.079 (panel b), a larger gain than any architectural modification we evaluated at matched capacity.

TABLE I
TRACE SCALING STUDY ON DATASET 4.

(a) Width-depth sweep (fixed $p=0.2$)

#	d_{model}	L	h	Params (M)	Mean r
1	128	2	4	0.9	0.582
2	192	2	4	1.8	0.591
3	256	2	4	3.1	0.601
4	256	4	4	5.9	0.612
5	1024	6	16	78.0	0.675

(b) Channel dropout sweep (fixed $d_{\text{model}}=1024$, $L=6$, $h=16$)

p	S1	S2	S3	Mean
0.1	0.617	0.544	0.774	0.645
0.2	0.680	0.569	0.777	0.675
0.3	0.688	0.588	0.784	0.687
0.4	0.716	0.605	0.793	0.705
0.5	0.731	0.624	0.775	0.710
0.6	0.776	0.622	0.773	0.724
0.7	0.772	0.620	0.768	0.720

Performance peaks at $p=0.6$ and declines at $p=0.7$. The decline at $p=0.7$ corresponds to only ~ 19 surviving channels per training step for S1 (62 electrodes), likely below the minimum needed to span the hand motor representation at ~ 1 cm inter-electrode spacing. Per-subject optima diverge slightly (S1 prefers $p=0.6$, S2 prefers $p=0.5$, S3 prefers $p=0.4$); the real constraint is that subjects with fewer channels (S2 at 48, Miller *wm* at 38) degrade faster with aggressive masking, consistent with a surviving-channel-count floor rather than a fixed optimal rate. A per-subject mixed oracle reaches 0.731, suggesting that per-patient dropout tuning is a low-cost deployment lever. Higher dropout rates also progressively delay the epoch of peak validation performance (from epoch 5 at $p=0.2$ to epoch 25+ at $p=0.6$), extending the window of productive gradient updates before overfitting.

C. Results

TRACE with $p=0.6$ channel dropout achieves a mean Pearson r of 0.724 ± 0.003 across the three Dataset 4 subjects (10 seeds), with per-subject scores of S1 0.762 ± 0.013 , S2 0.627 ± 0.006 , and S3 0.779 ± 0.005 , compared to FingerFlex (S1 0.575, S2 0.520, S3 0.692; mean 0.597) and our DTCNet [14] reproduction (S1 0.696, S2 0.598, S3 0.747; mean 0.680) under the same evaluation. On the nine-patient Miller finger-flexion subset, TRACE achieves a mean r of 0.508 compared to 0.436 for the U-Net baseline, with per-patient correlations ranging from 0.047 (*wm*, 38 channels) to 0.818 (*cc*, 63 channels), reflecting the wide variability in clinical grid coverage across epilepsy patients.

On Dataset 4, TRACE improves mean r by 0.127 over FingerFlex and 0.044 over DTCNet. The gains concentrate on S1 (+0.066 over DTCNet) while S2 improves by +0.029 and S3 is essentially flat at +0.032, consistent with the behavior of high channel dropout: subjects with denser and more redundant grids benefit most. On Miller data, absolute correlations drop relative to the competition corpus, expected given clinically placed grids, shorter recordings, and noisier

TABLE II
MILLER ECoG LIBRARY FINGER-FLEXION RESULTS, SORTED BY TRACE PERFORMANCE.

Patient	Ch.	FingerFlex (paper)	FingerFlex (ours)	TRACE
cc	63	0.71	0.725	0.818
jp	58	0.57	0.548	0.630
zt	61	0.63	0.540	0.602
bp	46	0.55	0.369	0.597
mv	43	0.56	0.460	0.548
jc	47	0.43	0.536	0.508
wc	64	0.42	0.383	0.449
ht	64	0.34	0.287	0.369
wm	38	0.22	0.083	0.047
Mean		0.49	0.436	0.508

labels, but TRACE improves the mean from 0.436 to 0.508 (+0.071), demonstrating that the attention-based advantage transfers to the sparse, suboptimal electrode coverage typical of real epilepsy-monitoring scenarios. Per-patient results on the nine-patient Miller finger-flexion subset, compared against both our U-Net reproduction and the FingerFlex paper-reported numbers [11], are given in Table II.

D. Reproducibility Audit

Table III consolidates published and reproduced Dataset 4 results. We reimplemented every baseline from the published description regardless of whether code was released.

FingerFlex [11], the only model with a public code release, achieved a mean r of 0.597 in our reproduction across multiple seeds, compared to the paper-reported 0.67. The gap reflects sensitivity to evaluation protocol details such as split policy and output smoothing, not implementation errors: rerunning FingerFlex under their published protocol ($\sigma=6$ smoothing, 100% of the official train split) yields a single-seed mean within ~ 0.02 of our default-protocol reproduction and still well below the 0.67 paper headline, consistent with best-of-seed selection in the original report rather than a missing component on our side. DTCNet [14] reproduced at 0.680, within 0.010 of the paper-reported 0.69, confirming it as a credible baseline and our primary convolutional target. DeepFingerNet [20] could not be reproduced despite five attempts with varying preprocessing interpretations and architecture fixes; our best result was 0.427, a gap of 0.26 from the claimed 0.69 and below our plain U-Net baseline. BC4D4 [18] is similarly excluded; its evaluation removes approximately 70% of samples via outlier filtering, making reproduction under standard protocols infeasible.

VI. DISCUSSION

A. Transformer Capacity and Decoding Performance

Transformer architectures address temporal modeling limitations through self-attention that can relate all token pairs within a window, enabling stronger long-context modeling than purely local convolutions [12]. Our scaling study (Table I) shows that TRACE benefits from capacity: mean r at fixed $p=0.2$ climbs from 0.582 at 0.9M parameters to 0.675 at

TABLE III
DATASET 4 FINGER DECODING. NUMBERS IN PARENTHESES ARE PAPER-CLAIMED; NUMBERS OUTSIDE ARE OUR REPRODUCTIONS.

Method	Architecture	S1	S2	S3	Mean
Liang & Bougrain (2012) [15]	Linear regression	0.45	0.39	0.59	0.48
Flamary & Rakotomamonjy (2012) [16]	Switching linear models	0.48	0.24	0.56	0.42
Yao et al. (2022) [24]	lightGBM	0.513	0.473	0.614	0.537
FingerFlex (2023) [11]	Wavelet + U-Net	0.575 (0.66)	0.520 (0.62)	0.692 (0.74)	0.597 (0.67)
DTCNet (2025) [14]	Dilated-transposed CNN	0.696 (0.71)	0.598 (0.59)	0.747 (0.77)	0.680 (0.69)
DeepFingerNet (2025) [20]	Nested U-Net	0.427 (0.71)	0.312 (0.59)	0.541 (0.77)	0.427 (0.69)
BC4D4 [†] (2025) [18]	Hybrid CNN-DNN	n/r (0.86)	n/r (0.81)	n/r (0.90)	n/r (0.86)
TRACE (2026)	Transformer	0.762	0.627	0.779	0.724

[†]Unreproducible; 70% of data removed by outlier filter.

the production configuration of ~ 78 M parameters. Separately, scaling channel dropout from $p=0.1$ to $p=0.6$ adds $+0.079$, a larger gain than any architectural modification at matched capacity. Both levers affect decoding quality substantially, though only capacity trades off against parameter count and training cost; channel dropout adds no overhead. This joint contribution is the central quantitative finding of this paper.

Relative to FingerFlex, which in our implementation occupies on the order of 10^5 trainable parameters, the full TRACE stack is several hundred times larger, before counting the quadratic memory cost of attention in long windows. Future work could quantify this memory cost in realistic operation settings. Attention-based decoding is thus easiest to justify when target hardware tolerates the footprint; for implant-adjacent or battery-limited pipelines, reducing model size while preserving the regularization benefit of aggressive channel dropout remains an open priority for TRACE and other transformer-based encoders.

B. Why Aggressive Channel Dropout Works in ECoG

Channel dropout functions as a form of forced distributed learning: by randomly masking majority subsets of electrodes during training, the model must extract motor information from cortical areas beyond the primary motor cortex (M1), including premotor, supplementary motor, and somatosensory regions, that carry weaker but complementary movement signals [7]. Without dropout, models converge to depend on a small cluster of dominant electrodes directly over M1, ignoring the distributed nature of the underlying cortical motor representation. Such aggressive input-level masking ($p=0.6$) is viable precisely because adjacent ECoG electrodes at ~ 1 cm spacing share substantial signal via volume conduction: dropping 60% of channels still preserves the majority of cortical information through neighboring survivors, unlike dropping 60% of learned features inside a network where each dimension may encode unique information. The observation that subjects with higher grid redundancy tolerate more aggressive dropout rates is consistent with this interpretation.

C. Patient Variability on the Miller ECoG Library

The Miller Library is substantially harder than Dataset 4 because electrode placement is dictated by clinical seizure

localization, not BCI-optimized targeting of hand sensorimotor cortex. Coverage of informative motor areas ranges from dense to marginal across patients, training data are shorter and less standardized, and finger-flexion labels are noisier than Dataset 4’s tightly cued paradigm. These factors cap the achievable correlation and inflate cross-subject variance: TRACE per-patient r spans 0.047–0.818, compared to 0.627–0.779 on Dataset 4. Despite this, TRACE raises the nine-patient mean from 0.436 to 0.508 ($+0.071$), with the largest gain on patient `bp` ($+0.228$). The two patients where TRACE underperforms the baseline (`jc`, -0.028 ; `wm`, -0.036) have among the fewest channels; `wm` in particular has only 38 electrodes, so $p=0.6$ channel dropout leaves roughly 15 active channels per step, likely too few for reliable decoding. Per-patient dropout tuning is a natural next step. This clinical heterogeneity is precisely the regime where channel dropout matters most: tolerating missing or uninformative electrodes is not an artificial regularization trick on Dataset 4 but a practical requirement for real epilepsy-monitoring grids.

D. Limitations

We frame decoding as single-task supervised regression of continuous flexion trajectories rather than joint training over heterogeneous control primitives; although the Miller Library spans multiple behaviors [19], harmonized multi-task objectives and shared heads across paradigms are not yet in scope. Although we report multi-seed means for TRACE on Dataset 4 (Section V-A), full multi-seed variance estimates for every baseline and for each Miller Library subject were not computed, so the reported Miller cohort mean $\pm$ std should be read as a cohort summary rather than an exhaustive robustness certificate. Finally, all models evaluated here share the same Morlet wavelet preprocessing pipeline; systematic ablation studies that isolate preprocessing choices from architectural contributions remain an open direction for the field.

VII. CONCLUSION

In this work, we address the under-explored use of transformer encoders for continuous ECoG motor decoding by proposing TRACE, which applies multi-head self-attention across electrode channels and time to capture spatiotemporal

structure that purely local convolutional receptive fields can miss. On BCI Competition IV Dataset 4, TRACE achieves a mean Pearson correlation of 0.724 ± 0.003 across subjects (10 seeds), improving on the FingerFlex convolutional baseline (0.597) and our DTCNet [14] reproduction (0.680). Ablation studies reveal that severe channel dropout ($p=0.6$), an input-level regularization strategy that exploits the spatial redundancy of ECoG grids, is the single most impactful design choice, contributing more than any architectural modification at matched capacity, a finding with direct practical implications for clinical deployment on variable-coverage grids. On the nine-patient Miller ECoG Library finger-flexion subset, where clinically placed grids make decoding harder, TRACE raises the mean from 0.436 to 0.508 over the U-Net baseline, demonstrating that the approach generalizes beyond a single optimized benchmark to the sparse electrode coverage of real clinical recordings.

Future work includes unifying training objectives and schedules across backbones, pairing TRACE with stronger statistical preprocessing pipelines, and reporting latency-aware metrics and memory usage alongside correlation. Clinical translation should not be measured by leaderboard increments alone, but by functional motor restoration in target populations. Realizing that shift will require coordinated advances in acquisition hardware, evaluation protocols, and clinical validation, together with interdisciplinary collaboration among machine learning researchers, neuroscientists, and practitioners.

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